Supplementary Material for "Observational Indistinguishability and Integrity Blind Regions Across the Evidence Boundaries of Hybrid Quantum-Classical Kernel Workflows"

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1 Purpose and Claim Boundary

This supplement records the complete formal statements with proofs, the experimental design, the adversary-model reading rule, the statistical units and their dependence structure, the validation checks, the full tables of the preregistered gates of artifacts 1.1.0 and 1.2.0, the secondary model-impact profile and the reproduction mapping behind the main article. It adds no QPU, calibrated-noise, scheduling, provider-security, multi-tenancy or operational Fleet Management evidence. The exact claims remain bounded to data-to-evaluation interventions, ideal-statevector quantum kernels, controlled binomial shot emulation and a local executable contract.

2 Formal Core with Proofs

2.1 Objects

A workflow state is $s = (X, P, C, K, E, f, y, R) \in \mathcal{S}$ with $\tilde{X} = P(X)$ and $\hat{y} = f(\tilde{X})$ indexed by item identity $i = 1, \ldots, n$. An intervention is a map $a : \mathcal{S} \to \mathcal{S}$, the baseline is s_0 and $s_a = a(s_0)$. Classes: T_y (label-only), $T_y^\pi \subset T_y$ (prior-preserving) and the feature-side mechanisms. The conclusion is $R(s)$ (balanced accuracy) and $\Delta_R(a) = R(s_0) - R(s_a)$; a is material at level τ if $|\Delta_R(a)| \geq \tau$. An information set $\mathcal{I}$ is a view $V_\mathcal{I} : \mathcal{S} \to \mathcal{O}_\mathcal{I}$; refinement $\mathcal{I} \sqsubseteq \mathcal{I}'$ means $V_\mathcal{I} = \pi \circ V_{\mathcal{I}'}$; the join $\mathcal{I} \vee \mathcal{W}$ is the pair of views. A reference $\rho = V_\mathcal{J}(s_0)$ is trusted when no intervention of the class can alter it and item-aligned when $V_\mathcal{J}$ is item-indexed.

2.2 Statements and Proofs

Lemma 1 (structural blindness). If $V_\mathcal{I}(s_a) = V_\mathcal{I}(s_0)$ then $S(s_a) = S(s_0)$ for every sensor S admissible in $\mathcal{I}$.

Proof: $S(s_a) = S(V_\mathcal{I}(s_a)) = S(V_\mathcal{I}(s_0)) = S(s_0)$. For a randomized sensor whose seed is part of the view the identity holds path-wise; with a fresh seed the two outputs are equal in distribution. □

Definition (blind regions and gaps). $B_\mathcal{I}(\mathcal{A}) = \{a \in \mathcal{A} : V_\mathcal{I}(a(s_0)) = V_\mathcal{I}(s_0)\}$; $B_{\mathcal{I},\mathcal{S}}(\mathcal{A}) = \{a : S(a(s_0)) = S(s_0)\ \forall S \in \mathcal{S}\}$; $G_\mathcal{I}(\tau) = \{a : |\Delta_R(a)| \geq \tau, a \in B_\mathcal{I}\}$ and $G_{\mathcal{I},\mathcal{S}}(\tau)$ analogously. By Lemma 1, $B_\mathcal{I} \subseteq B_{\mathcal{I},\mathcal{S}}$ and $G_\mathcal{I} \subseteq G_{\mathcal{I},\mathcal{S}}$, with equality iff $\mathcal{S}$ is baseline-separating on the orbit of $\mathcal{A}$: it takes a different value at $a(s_0)$ than at s_0 whenever the views differ. Pairwise separation of intervened states (injectivity on the orbit), which identification of the intervention would need, is never assumed. A reference $\rho = V_\mathcal{J}(s_0)$ is an *aggregate* reference (level B) when $V_\mathcal{J}$ is a permutation-invariant summary of the batch such as the confusion matrix M, the label histogram or R, and *item-aligned* (level C) when it is an item-indexed tuple.

Proposition 1 (monotonicity). If $\mathcal{I} \sqsubseteq \mathcal{I}'$ then $B_{\mathcal{I}'}(\mathcal{A}) \subseteq B_\mathcal{I}(\mathcal{A})$ and $G_{\mathcal{I}'}(\tau) \subseteq G_\mathcal{I}(\tau)$.

Proof: If $V_{\mathcal{I}'}(s_a) = V_{\mathcal{I}'}(s_0)$ then $V_\mathcal{I}(s_a) = \pi(V_{\mathcal{I}'}(s_a)) = \pi(V_{\mathcal{I}'}(s_0)) = V_\mathcal{I}(s_0)$. Membership in the gap adds only the materiality condition, which does not depend on the view. □

Corollary 1. With f fixed and deterministic: (a) $T_y \subseteq B_{\mathcal{I}_{XF}} \subseteq B_{\mathcal{I}_X}$ and every sensor admissible in $\mathcal{I}_X$ or $\mathcal{I}_{XF}$ is invariant under a label-only change; (b) $T_y^\pi \subseteq B_{\mathcal{I}_{Y_m}}$ and every sensor admissible in $\mathcal{I}_{Y_m}$ is invariant under a prior-preserving change.

Proof: (a) T_y fixes X, P and f, hence $\tilde{X}$ and $\hat{y} = f(\tilde{X})$; the multiset of pairs $(\tilde{x}_i, \hat{y}_i)$ is unchanged. Apply Lemma 1 and Proposition 1 with $\mathcal{I}_X \sqsubseteq \mathcal{I}_{XF}$. (b) T_y^π preserves the class histogram by definition. □

Proposition 2 (closure). $B_{\mathcal{I} \vee \mathcal{W}}(\mathcal{A}) = B_\mathcal{I}(\mathcal{A}) \cap B_\mathcal{W}(\mathcal{A})$; a class $\mathcal{A} \subseteq B_\mathcal{I}$ is completely separable in $\mathcal{I} \vee \mathcal{W}$ iff $V_\mathcal{W}(a(s_0)) \neq V_\mathcal{W}(s_0)$ for every $a \in \mathcal{A}$.

Proof: The joint view equals the baseline joint view iff both components do. Necessity: an a that leaves $V_\mathcal{W}$ unchanged stays in the joint blind region. Sufficiency: a changed component changes the pair. □

Corollary 2. With f fixed and deterministic, every view factoring through $(\tilde{X}, f(\tilde{X}), \mathrm{hist}(y))$ leaves every $a \in T_y^\pi$ in its blind region. The multiset $\mathcal{I}_{XFY}$ separates $a \in T_y^\pi$ unless a permutes labels only among items with identical $(\tilde{x}, \hat{y})$; an item-aligned reference $\mathcal{I}_{XFY}^\star$ separates every non-identity relabeling.

Proof: Apply Proposition 2 with $\mathcal{W}$ the view in question: by Corollary 1 each of the three components is invariant, so their join is. For the multiset of triples, two states have the same multiset iff a permutation of items maps one to the other; since $(\tilde{x}_i, \hat{y}_i)$ is fixed, such a permutation can only move labels among items with identical $(\tilde{x}, \hat{y})$. An item-indexed tuple has no such freedom. □

Proposition 3 (materiality forces aggregate separability). If $R = g(M)$ with M the confusion matrix, $a \in T_y$ and $\Delta_R(a) \neq 0$, then $M(s_a) \neq M(s_0)$; hence a is separable in the confusion view and in $\mathcal{I}_{XFY}$, and $G_{\mathcal{I}_{XFY}}(\tau) \cap T_y = \emptyset$ for $\tau > 0$.

Proof: $M(s_a) = M(s_0)$ implies $R(s_a) = g(M(s_a)) = g(M(s_0)) = R(s_0)$. The confusion matrix is a function of the multiset of triples, so a changed M implies a changed multiset. □

Proposition 4 (separability and anchoring). (i) For any auditor whose decision is a function of $V_{\mathcal{I}}(s)$ (including a seed carried in the view), $a \in B_{\mathcal{I}}$ implies $\mathrm{fire}(s_a) = \mathrm{fire}(s_0)$. (ii) A reference-anchored auditor with trusted item-aligned $\rho = V_{\mathcal{J}}(s_0)$ and rule "fire iff $d(V_{\mathcal{J}}(s), \rho) > 0$" has zero false-alarm probability and detects every $a \notin B_{\mathcal{J}}$ with probability one.

Proof: (i) is Lemma 1 applied to the decision function. (ii) $d(\rho, \rho) = 0$ and $d(V_{\mathcal{J}}(s_a), \rho) > 0$ iff $V_{\mathcal{J}}(s_a) \neq \rho$, which is $a \notin B_{\mathcal{J}}$ because ρ is unaltered. □

Corollary 3 (which reference certifies which integrity). Let $a \in T_y$ with f fixed and deterministic. (a) With a trusted aggregate reference $M_0 = M(s_0)$ of the same batch, "fire iff $M(s_a) \neq M_0$" detects every violation of aggregate integrity and, by Proposition 3, every material a; a trusted R_0 detects every material a and nothing else. (b) Any reference that factors through M, $\mathrm{hist}(y)$ or R is invariant under relabelings that permute labels among items with equal predictions (counterexample C1), which violate item-identity integrity; an item-aligned reference y_0 detects every non-identity relabeling.

Proof: (a) Proposition 4(ii) with $\mathcal{J}$ the confusion view, then Proposition 3. (b) Swapping y_i, y_j with $\hat{y}_i = \hat{y}_j$ moves one item from cell $(\hat{y}_i, y_i)$ to $(\hat{y}_i, y_j)$ and another from $(\hat{y}_j, y_j)$ to $(\hat{y}_j, y_i)$; the moves cancel in M, hence in every function of M, and the histogram is unchanged, whereas the item-indexed tuple differs in positions i and j. □

Proposition 5 (union of calibrated sensors versus family calibration). (a) Let $E_1, \ldots, E_m$ be the null firing events of m sensors with $p_j = \Pr_0[E_j] \leq \alpha$. Then

$$\max_j p_j \;\leq\; \Pr_0\Big[\bigcup_j E_j\Big] \;\leq\; \min\Big(1, \sum_j p_j\Big) \;\leq\; \min(1, m\alpha). \tag{1}$$

Without assuming that some $p_j = \alpha$ the universal lower bound is $\max_j p_j$, which can be 0. If every $p_j = \alpha$, then $\alpha \leq \Pr_0[\bigcup_j E_j] \leq \min(1, m\alpha)$: the lower bound is attained by nested (coincident) events, the upper bound $m\alpha$ (when $m\alpha \leq 1$) by mutually disjoint events, and under independence $\Pr_0[\bigcup_j E_j] = 1 - \prod_j(1 - p_j) = 1 - (1 - \alpha)^m$, strictly between the two bounds for $m \geq 2$ and $0 < \alpha < 1$. (b) Let $v_1, \ldots, v_n$ be the sensor vectors of the calibration draws and v_{n+1} that of the audited batch. For a family F define the score of any vector v as $U(v) = \max_{j \in F} \#\{i \leq n : v_{j,i} < v_j\}/n$ and let q be the k-th smallest of $U(v_1), \ldots, U(v_n)$ with $k = \lceil (n+1)(1-\alpha) \rceil$. If the $n+1$ draws are exchangeable, then

$$\Pr_0[U(v_{n+1}) > q] \;\leq\; \frac{n+2-k}{n+1} \;\leq\; \alpha + \frac{1}{n+1}. \tag{2}$$

For $n = 200$ and $\alpha = 0.05$, $k = 191$ and the bound is $11/201 = 0.0547$. For a single sensor the rule coincides with "fire iff v_{n+1} exceeds the k-th smallest calibration value", whose exact exchangeable level is $(n+1-k)/(n+1) = 10/201 = 0.0498$. (c) Gate F uses (b) with $n = 200$ per (environment, dimension, model) cell and evaluates it on 200 disjoint evaluation draws; the 20 draws of a run overlap and the two halves are different rows of one staged table, so the premise does not hold exactly, and the observed rates (Table 14) measure that deviation rather than confirm the bound.

Proof: (a) The union contains each E_j (lower bound); Boole's inequality gives the upper bound; nested, disjoint and independent events attain or realise the stated values, the last by inclusion–exclusion. (b) Let $\tilde{U}_i$ be the score of draw i computed against the *other* n draws of the augmented set $\{1, \ldots, n+1\}$. Under exchangeability the $\tilde{U}_i$ are exchangeable, so the rank of $\tilde{U}_{n+1}$ among the $n+1$ scores is uniform up to ties and $\Pr[\tilde{U}_{n+1} \geq \tilde{U}_{(k)}] \leq (n+2-k)/(n+1)$, where $\tilde{U}_{(k)}$ is the k-th smallest of $\tilde{U}_1, \ldots, \tilde{U}_n$; ties count against firing and only lower the probability. Now $U(v_{n+1}) = \tilde{U}_{n+1}$ and, for $i \leq n$, $U(v_i) \leq \tilde{U}_i \leq U(v_i) + 1/n$, because the augmented comparison set adds one element; hence $q \geq \tilde{U}_{(k)} - 1/n$ and, on the grid $\{0, 1/n, \ldots, 1\}$, $\{U(v_{n+1}) > q\} \subseteq \{\tilde{U}_{n+1} \geq \tilde{U}_{(k)}\}$. For a single sensor $U(v_i) = \tilde{U}_i$ exactly and the level is $(n+1-k)/(n+1)$. □

Proposition 6 (no local authentication without an uncontrolled root). Let a *local verifier* be any map $\mathrm{acc} : \mathcal{O}_{\mathcal{I}} \times \mathrm{Ref} \to \{\text{accept}, \text{reject}\}$ of the observed view and a reference bundle, let $\mathcal{S}_H$ be the honest family of states and $\rho_H(s) = V_{\mathcal{J}}(s)$ the reference the honest process attaches to s. Assume (honest completeness) $\mathrm{acc}(V_{\mathcal{I}}(s), \rho_H(s)) = \text{accept}$ for every $s \in \mathcal{S}_H$, and (joint control) the adversarial class can produce $(V_{\mathcal{I}}(s'), \rho_H(s'))$ for some $s' \in \mathcal{S}_H$ with $s' \not\sim_{\mathcal{I}} s_0$. Then the verifier accepts the substituted pair, which is observationally an honest run of s', so no function of that input establishes authenticity of the served state relative to s_0; if $R(s') \neq R(s_0)$ a material change is served. Conversely, if a component $\rho^\star = V_{\mathcal{K}}(s_0)$ of the bundle cannot be written by the class, "reject iff $V_{\mathcal{K}}(s) \neq \rho^\star$" rejects every substitution with $V_{\mathcal{K}}(s') \neq V_{\mathcal{K}}(s_0)$.

Proof: The substituted input equals the honest input of s'; by honest completeness it is accepted, and acceptance is the same event as for an honest run of s', so no function of the input distinguishes the two. The converse is Proposition 4(ii) applied to $\mathcal{K}$. The statement concerns authenticity, not consistency: by Proposition 3 a material substitution changes the joint view, but the change is invisible to a verifier all of whose evidence the class can rewrite. □

2.3 Counterexamples

Table 1 restates the witnesses of the main text. C1 is the swap of two labels between items with equal predictions; C3 between items with different predictions; C4 uses two compensating flips that move the same number of items across the diagonal of M in each class; C5 and C6 are

TABLE 1
Counterexample witnesses realised in the frozen expansion (count / denominator). W1–W4 are evaluation-label rows, W5–W6 feature-side rows, W7 all intervened rows, W8 material label rows; W9 and W10 count the label rows that a trusted aggregate reference, respectively only an item-aligned reference, detects (Corollary 3).

ID	Property	Count
W1	identity change without joint-outcome or conclusion change	808 / 3,600
W2	conclusion change without model-output change	2,617 / 3,600
W3	marginal invariance with conclusion change	1,059 / 1,800
W4	confusion-profile change without conclusion change	175 / 3,600
W5	model-output change without conclusion change	296 / 7,200
W6	representation change without output change	2,513 / 7,200
W7	apparent improvement under intervention	1,534 / 10,800
W8	material conclusion change always changes the confusion profile (label path)	2,617 / 2,617
W9	label rows detectable by a trusted aggregate (confusion-matrix) reference of the same batch	2,792 / 3,600
W10	label rows detectable only by an item-aligned reference	808 / 3,600

feature-side; C7 counts every intervened row whose balanced accuracy increased; C8 is Proposition 3; W9 and W10 count the label rows that a trusted aggregate reference of the same batch detects (every material one) and those that only an item-aligned reference exposes (Corollary 3).

3 Adversary Model: Reading Rule

Every executed intervention is assigned to one of four classes: *fault robustness* (benign or accidental change), *integrity corruption* (non-adaptive change by a party that does not model the auditor), *adversarial attack* (targeted change to alter the reported conclusion) and *adaptive attacker* (targeted change designed with knowledge of the monitored statistics). The complete per-class record (actor, capability, access point, knowledge, objective, budget, alterable assets, roots the class cannot write, observable information set and supported claim) is `manuscript/ADVERSARY_MODEL.md` in the artifact; the main article prints the summary table. The trusted computing base is: the local runtime and host; the training data, training routine and fitted model; the reference input array and preprocessing declaration; the reference circuit and probe kernels; SHA-256 collision resistance; the item-level label reference only where $\mathcal{I}^{\star}_{XFY}$ is declared; and the code that builds and verifies the envelope.

4 Frozen Experimental Design

4.1 Expansion Environments

The expansion comprises eight fixed environments (Table 2). "OOD" means a prespecified source-to-target construction, not a sampled deployment population. Every environment uses five split seeds and two nested model seeds and evaluates projected dimensions 8, 10 and 12. The CICIDS scale gate uses 256/256 train/evaluation samples; all others use 128/128.

Each staged table contains 3,000 balanced binary rows before experimental subsampling. Numeric non-finite values are replaced during staging; categorical fields are excluded. Several temporal targets yield clean performance near chance, limiting the impact that later interventions can express; the exact information-set invariances do not depend on clean headroom.

4.2 Perturbation Suite

The paper-core suite crosses six mechanisms with nominal strengths 0.02, 0.05 and 0.10 (Table 3). Artifact identifiers retain the historical family string `target_shift`; the scientific text uses *evaluation-label intervention* because these rows do not instantiate the statistical label-shift problem.

5 Information-Set Coverage Contract

Table 4 records what each evidence regime can establish in the implemented study. Structural blindness means that the view is unchanged by construction; it is stronger and narrower than observing zero empirical response in a finite sample, and both are distinct from calibrated detectability.

Every machine-readable coverage record contains seven fields: workflow boundary; protected asset; adversary or failure class including fixed elements; available information regime and trusted references; sensor, threshold and decision-level false-alarm budget; covered perturbations and structural and calibrated residual blind regions; response or countermeasure.

6 Deduplication and Statistical Units

Gate 1 emits 7,980 raw model-intervention rows; different quantum-map jobs repeat the same classical SVC pipeline, producing 3,420 repeated rows that are accepted as duplicates only after numerical identity checks, leaving 4,560 unique observations. The expansion emits 13,680 raw rows from 360 completed CSV/JSON job pairs with 2,280 verified repeats, leaving 11,400 unique observations. The strict builder fails if a job is missing, an environment contains an unexpected design, or a duplicated row disagrees.

Coverage counts use prespecified cells as denominators. For the secondary suite-average model profile, nested model-seed values are averaged inside each of five split seeds; the split seed is the inferential unit and the interval is $\bar{d} \pm t_{0.975,4}\, s_d/\sqrt{5}$. For the calibration gates the dependence structure is explicit: each of the 240 runs (eight environments $\times$ five split seeds $\times$ two model seeds $\times$ three dimensions) draws 20 calibration and 20 evaluation batches as overlapping simple random subsets of one pool half, and the 200 draws of a cell come from ten pool halves; the inferential unit is the (environment, split-seed) cluster (40 clusters, five per environment). Pooled rates over draws and the Clopper–Pearson intervals of artifact 1.1.0 are descriptive. No multiplicity adjustment or population transport is claimed.

7 Exact-Statevector Engine Validation

The expansion evaluator replaces pairwise compute-uncompute execution with a mathematically equivalent ideal-statevector calculation: it prepares each unique state $|\phi(x)\rangle$ once and forms $K(x,y) = |\langle\phi(x) \mid \phi(y)\rangle|^2$ by matrix multiplication, with the same eigenvalue-clipping PSD repair as the frozen Qiskit reference. Validation has four layers: symmetric and cross-kernel unit matrices agree with the reference at relative and absolute tolerance 10^{-10}; symmetry, unit diagonal, numerical range, cache bounds and cache reuse are unit-tested; a CICIDS 128/128 end-to-end replay (split/model seed 42, dimension 8, ZZ, paper-core suite)

TABLE 2
Prespecified expansion environments.

ID	Environment	Training source	Evaluation source	Samples	Profiles
E1	CICIDS scale/ID	Frozen CICIDS subset	Same staged source	256/256	SVC, ZZ
E2	UNSW-NB15 ID	Balanced staged UNSW	Same staged source	128/128	SVC, ZZ, PauliXYZ, Z/PauliXZ-equivalent
E3	ToN-IoT ID	Balanced staged ToN-IoT	Same staged source	128/128	SVC, ZZ, PauliXYZ, Z/PauliXZ-equivalent
E4	CICIDS Tue→Wed	Tuesday traffic	Wednesday traffic	128/128	SVC, ZZ
E5	CICIDS Tue→Fri-PortScan	Tuesday traffic	Friday PortScan traffic	128/128	SVC, ZZ
E6	CICIDS Wed→Thu-Web	Wednesday traffic	Thursday Web traffic	128/128	SVC, ZZ
E7	CICIDS Wed→Fri-AM	Wednesday traffic	Friday morning traffic	128/128	SVC, ZZ
E8	UNSW temporal OOD	Prespecified UNSW source period	Prespecified UNSW target period	128/128	SVC, ZZ

TABLE 3
Frozen 18-condition intervention suite.

IDs	Mechanism	Strengths
1–3	Feature sign flip	0.02, 0.05, 0.10
4–6	Feature-wise mean shift	0.02, 0.05, 0.10
7–9	Scaling drift	0.02, 0.05, 0.10
10–12	Feature dropout	0.02, 0.05, 0.10
13–15	Prior-preserving evaluation-label flip	0.02, 0.05, 0.10
16–18	Random evaluation-label flip	0.02, 0.05, 0.10

preserves every categorical, discrete performance, impact and integrity-signal endpoint, with only ROC-AUC and its derived drops differing by at most 1.221896×10^{-4} (one half-tie increment for 64 positive and 64 negative examples); and the prespecified 256/256 pilot completes. The engine is tagged `exact_statevector`; it supplies no finite-shot, noise-model or hardware evidence.

8 Quantum Gate: Conditions and Coverage

The quantum gate crosses five split seeds, dimensions 4/6/8 and 11 conditions, giving 165 cells. Table 5 reports response fractions over the 15 split-dimension cells per condition. A value of one means non-zero raw response in all 15 cells at the frozen tolerance; it is not calibrated power.

All nine frozen acceptance checks pass: 165 cells are present; clean evidence is invariant; benign transpilation and the common-unitary rewrite alter provenance but preserve semantic kernels and outputs; every data-dependent circuit mutation changes the semantic kernel; symmetry and diagonal sensors catch their respective malformed edits; the PSD-preserving mixture passes algebra but is exposed by trusted provenance/semantics; and both shot estimators produce non-zero repeat discrepancy in every cell. The shot conditions sample exact fidelities: they exercise the contract under controlled stochastic estimation but are neither sampler-backend nor QPU runs. PSD repair is containment, not proof of integrity.

9 Executable Contract and Policy Composition

The prototype evaluates four ordered contracts: input/preprocessing (input array and preprocessing declaration hashes; mismatch blocks); circuit/kernel (circuit and kernel provenance, semantic probes, symmetry, diagonal, eigenvalue and PSD-repair evidence; an equivalent rewrite passes only if explicitly approved); execution/result (declared execution metadata, repeated-estimation discrepancy and prediction provenance; approved stochastic execution is held); evaluation/report (item-level labels bound to predictions and balanced accuracy recomputed; prior-preserving corruption blocks even when the marginal is identical). Contracts form an ordered SHA-256 chain: later modification is detectable, but the chain is not signed and does not authenticate a provider. All passes yield `allow`; at least one review and no failure yields `hold`; any failure yields `block`. Six prespecified scenarios and all six chains pass eight strict acceptance checks in the frozen artifact. The policy layer of artifact 1.2.0 composes with these contracts by the maximum in the lattice; the composition over the six frozen envelopes is Table 6, where an exact violation is derived from the recorded evidence with the approval flags respected.

10 Preregistered Reinforcement Gates (Artifact 1.1.0)

Three sensitivity gates were added after the 1.0.0 evidence was frozen. Their protocol, endpoints and one implementation clarification (amendment A1) were recorded in `manuscript/paper15_v11_reinforcement_prereg.md` before execution; results are reported as executed. The gates reuse the exact-statevector engine, the frozen split and model seeds and the frozen 18-condition suite.

10.1 Gate N: Null Calibration with Disjoint Clean Pools

For each of the 240 expansion runs the clean pool is the set of evaluation-side rows not selected into the frozen evaluation batch (900 $- n$ in ID protocols, 3,000 $- n$ in OOD protocols), transformed with the same training-fitted projection and scaler as the frozen run. A permutation seeded by the split

TABLE 4
Evidence regimes, implemented sensors, demonstrated blind regions and calibrated status.

Regime	Evidence	Implemented sensor family	Structural blind region	Calibrated / exact status
$\mathcal{I}_X$	Features	Feature JSD, MMD, KS rejection	Every label-only change	Family-calibrated decision FPR 0.061; label path 0/3,600
$\mathcal{I}_{XF}$	Features, scores, predictions	$\mathcal{I}_X$ plus score JSD, prediction-rate shift, prediction JSD	Every label-only change for a fixed predictor	Family-calibrated FPR 0.073; label path 0/3,600
$\mathcal{I}_{Y_m}$	Label class counts	Prior delta and label JSD	Prior-preserving relabeling; every feature-side change	FPR 0.048; contains nothing in this suite
$\mathcal{I}_{XFY}$ (batch)	Item triples, no reference	Confusion-profile statistics added	None for material label changes (Proposition 3) but statistically undetectable: 16–43 of 2,617	Family-calibrated FPR 0.079
$\mathcal{I}^{\star}_{XFY}$	Same items against a trusted reference	Prediction disagreement, confusion-profile deltas, item-level label mismatch (exact-zero null)	None	Exact: 2,617/2,617 material label rows; 0 unsafe allows; 0 false holds
$\mathcal{I}_Q$	Circuit, parameters, kernel, estimation evidence	Hashes, semantic probes, algebra, repeated estimation, output changes	Composition-dependent: hashes overreact to equivalence; algebra misses PSD-preserving substitution; outputs miss sub-decision changes	165/165 cells; nine acceptance checks

TABLE 5
Quantum-gate sensor response fractions.

Condition	Circuit prov.	Kernel prov.	Semantic kernel	Algebraic kernel	Repeated estimate	Output change
Clean	0.00	0.00	0.00	0.00	0.00	0.00
Benign transpilation	1.00	1.00	0.00	0.00	0.00	0.00
Common-unitary rewrite	1.00	1.00	0.00	0.00	0.00	0.00
Parameterized RZ 0.02	1.00	1.00	1.00	0.00	0.00	0.00
Parameterized RZ 0.10	1.00	1.00	1.00	0.00	0.00	0.13
Feature-map reps 1→2	1.00	1.00	1.00	0.00	0.00	0.73
Asymmetric kernel edit	0.00	1.00	1.00	1.00	0.00	0.00
Kernel diagonal erosion	0.00	1.00	1.00	1.00	0.00	0.00
PSD-preserving kernel mix	0.00	1.00	1.00	0.00	0.00	0.07
Shot estimator 256	0.00	1.00	1.00	1.00	1.00	0.47
Shot estimator 1,024	0.00	1.00	1.00	1.00	1.00	0.20

TABLE 6
Composition of the frozen four-contract envelopes with the policy layer under $\mathcal{I}^{\star}_{XFY}$ (lattice maximum). Approved rewrites and approved stochastic estimation are not exact violations; the contract's own `hold` for the approved emulator is preserved.

Scenario	Contract	Exact viol.	Policy	Composed
Clean	`allow`	no	`allow`	`allow`
Approved transpilation	`allow`	no	`allow`	`allow`
Parameterized circuit mutation	`block`	yes	`block`	`block`
PSD-preserving kernel substitution	`block`	yes	`block`	`block`
Prior-preserving label corruption	`block`	yes	`block`	`block`
Approved 256-shot emulator	`hold`	no	`allow`	`hold`

seed divides the pool into a calibration half and an evaluation half; twenty clean batches of n rows are drawn from each half by simple random sampling without replacement; the suite also includes an identity control and two near-null controls (Gaussian noise $\sigma = 0.001$, scaling $\alpha = 0.001$) applied in place. Ten distributional sensors are calibrated per (environment, dimension, model) cell at the 191st of 200 pooled calibration values ($\alpha = 0.05$ per sensor; fire if strictly greater); thresholds are frozen before any attacked row is examined; the empirical false-alarm rate is the firing fraction on the 200 disjoint evaluation draws. Two auditors are distinguished (amendment A1): the batch-level auditor uses only calibrated distributional sensors, and the item-aligned auditor compares the same items against trusted references with an exact-zero null. In E1 ($n = 256$) each pool half holds 322 rows, so the draws overlap almost entirely and E1 rates are indicative only.

Results. All 240 runs completed (14/14 acceptance checks). Table 7 reports the pooled per-sensor false-alarm rates (0.029–0.069 against the nominal 0.05) and the uncorrected regime unions; Table 8 the environment dependence; Table 9 the calibrated detection rates and the near-null control responses; Table 10 the label path (calibrated $\mathcal{I}_X$, $\mathcal{I}_{XF}$ and prior-preserving $\mathcal{I}_{Y_m}$ fire on none of the 3,600 label observations; the batch-level $\mathcal{I}_{XFY}$ union fires on 43; the item-aligned auditor detects all 3,600); and Table 11 the cluster structure that makes the KS sensor react to in-place perturbations of order 10^{-3} standard deviations while its between-batch null stays narrow. The intervals in Table 7 are descriptive.

10.2 Gates P and T: Preprocessing Ablation and Tuned Baselines

The CICIDS ID design (128/128, paper-core, ZZ and SVC, five splits, two model seeds, three dimensions) is re-executed under three scaler configurations (P0 reference: standard for SVC, min–max to 2π for QSVC; PA standard for both; PB min–max for both) and with both learners tuned on training rows only (SVC over $C \in \{0.1, 1, 10, 100\} \times \gamma \in$

TABLE 7
Gate N: empirical false-alarm rate of the null-calibrated sensors and batch-level regimes on 12,000 disjoint clean evaluation draws (all eight environments pooled; nominal $\alpha = 0.05$ per sensor). Intervals are descriptive Clopper–Pearson 95% intervals: draws within a cell share a pool half and are not independent. Regime rows are unions without correction.

Sensor / regime	Fires	Rate	95% CI
Feature JSD	600	0.050	[0.046, 0.054]
Feature MMD	822	0.069	[0.064, 0.073]
KS rejection rate	342	0.029	[0.026, 0.032]
Score JSD	668	0.056	[0.052, 0.060]
Prediction-rate shift	520	0.043	[0.040, 0.047]
Prediction JSD	586	0.049	[0.045, 0.053]
Label prior shift	570	0.048	[0.044, 0.051]
Label JSD	570	0.048	[0.044, 0.051]
Confusion profile L_1	517	0.043	[0.040, 0.047]
Confusion profile JSD	638	0.053	[0.049, 0.057]
$\mathcal{I}_X$ (any sensor)	1500	0.125	[0.119, 0.131]
$\mathcal{I}_{XF}$ (any sensor)	2438	0.203	[0.196, 0.210]
$\mathcal{I}_{Y_m}$ (any sensor)	570	0.048	[0.044, 0.051]
$\mathcal{I}_{XFY}$ (any sensor)	3113	0.259	[0.252, 0.267]

$\{\text{scale}, 0.01, 0.1, 1\}$, QSVC over $C \in \{0.1, 1, 10, 100\}$ on the precomputed training kernel, stratified five-fold cross-validation, balanced-accuracy scoring, ties toward the smallest C) in CICIDS ID and in the OOD environment with the highest mean clean SVC balanced accuracy in the frozen tables (UNSW temporal OOD, E8, 0.753). Table 12 reports the paired differences and Table 13 the clean balanced accuracies and selected constants. The direction of the secondary profile is stable (five of five positive split clusters in every configuration, dimension and environment); its magnitude is conditional: standardizing the quantum branch reduces the difference to roughly a third, matched min–max scaling leaves it unchanged, tuning widens it in CICIDS ID as ZZ gains clean headroom and leaves UNSW temporal OOD unchanged. Exact label-path invariance holds in every configuration.

11 Preregistered Policy Gates (Artifact 1.2.0)

Two gates were preregistered on 2026-09-07 in `manuscript/paper15_v12_policy_prereg.md` before any analysis and computed from the frozen 1.1.1 outputs without re-executing any kernel or draw. Amendment A1 records the finding that the 1.1.x column `impact_bal_acc` is the positive part of the signed conclusion change, so that the material label-path count under the prespecified $|\Delta_R| > 0$ definition is 2,617 (2,184 lowered, 433 raised) rather than the 2,184 lowered rows alone; check D3 was corrected accordingly and nothing else changed.

11.1 Gate F: Family-Wise Calibration

For every cell and regime the family score of a batch is the maximum over the regime's sensors of the fraction of calibration values strictly below the observed value, and the family threshold is the 191st smallest calibration family score; the regime fires when the score exceeds it. The frozen per-sensor thresholds are reproduced from the calibration draws as a consistency check (F0). Table 14 gives the decision-level false-alarm rates of the union and family rules, Table 15 the per-environment cluster means with five-cluster t intervals, and Table 16 the detection rates on the frozen interventions and the near-null controls. The family-calibrated $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ rules fire on none of the 3,600 label observations and the $\mathcal{I}_{Y_m}$ rule on none of the 1,800 prior-preserving ones (F3); the batch-level $\mathcal{I}_{XFY}$ family rule fires on 16 of the 2,617 material label rows.

11.2 Gate D: End-to-End Decisions

Four policies (P0 serve-always; P1 the uncalibrated union of per-sensor rules; P2 the family-calibrated rule; P3 the strict variant that holds whenever a protected boundary is neither exactly nor statistically covered) are evaluated over five regimes on 12,000 clean evaluation draws (the 1,200 exact-zero rows for $\mathcal{I}^{\star}_{XFY}$), 1,200 near-null shams and the 10,800 frozen intervened observations, of which 7,008 are material at $|\Delta_R| > 0$ (1,870 at $\tau = 0.05$). `block` is reserved for exact violations against a trusted reference. Table 17 gives the complete decision counts, Table 18 the materiality sensitivity and Table 19 the unsafe allows by mechanism under P2. The five consistency checks D1–D5 (serve-always allows every material observation; structural blindness is reproduced by the decisions; the trusted regime blocks every material label row and every exact violation; decisions are total and deterministic; the composition reproduces the frozen contract actions) all pass.

12 Secondary Model-Impact Profile

The ZZ fidelity kernel has the largest mean conclusion impact in the frozen Gate-1 ID suite. Relative to SVC-RBF, mean impact ratios are 2.524, 3.094 and 3.323 at dimensions 8, 10 and 12; paired within-split differences are 0.02427 (95% CI 0.01200–0.03654), 0.03289 (0.01998–0.04580) and 0.03855 (0.02773–0.04937), with all five split clusters positive at every dimension (ZZ/SVC impacts 0.04019/0.01592, 0.04860/0.01571 and 0.05514/0.01659). Across the eight expansion environments the mean ZZ-minus-SVC impact is positive in 7/8 environments at dimension 8 and 8/8 at dimensions 10 and 12; environment-specific intervals exclude zero in 4/8, 7/8 and 5/8; means range from -0.00009 to 0.09382, 0.00107 to 0.09508 and 0.00403 to 0.11585 (Fig. 1). Clean balanced accuracy and impact correlate positively across expansion cells ($r = 0.714$, exploratory). No pooled population effect is claimed, and the conditional-auditability result does not depend on this ordering. Figure 2 shows the raw (uncalibrated) sensor response fractions by regime that motivated the calibration gates.

13 Reproduction and Evidence Manifest

The recommended read-only verification is

```
python -m src.experiments.verify_publication_artifact
       --root publication/artifact
```

It verifies the artifact-wide SHA-256 manifest, the six embedded evidence manifests and their 65 outputs, row counts, the primary-claim counts including the signed label-path counts, the calibrated label-path consistency checks of the reinforcement gate and the policy-level claims of the 1.2.0 gates. The complete test command is

```
python -m pytest tests -q -p no:cacheprovider
```

TABLE 8
Gate N: per-environment false-alarm rates on 1,200 evaluation draws (three dimensions, two or four models, 200 draws per cell). Sensor range is the minimum and maximum per-sensor rate in the environment; regime columns are batch-level unions. E1 draws 256 rows from 322-row pool halves and is indicative only.

Environment	n	Pool half	Sensor range	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{Y_m}$	$\mathcal{I}_{XFY}$
E1 CICIDS ID 256	256	322	0.023–0.130	0.203	0.245	0.045	0.331
E2 UNSW-NB15 ID	128	386	0.008–0.070	0.063	0.135	0.070	0.210
E3 ToN-IoT ID	128	386	0.025–0.172	0.215	0.280	0.070	0.358
E4 Tue→Wed	128	1436	0.025–0.082	0.150	0.230	0.025	0.254
E5 Tue→Fri-PortScan	128	1436	0.018–0.051	0.066	0.160	0.025	0.198
E6 Wed→Thu-Web	128	1436	0.013–0.057	0.070	0.163	0.025	0.181
E7 Wed→Fri-AM	128	1436	0.025–0.066	0.133	0.212	0.025	0.258
E8 UNSW temporal OOD	128	1436	0.032–0.092	0.072	0.192	0.050	0.237

TABLE 9
Gate N: detection rate of each regime on the frozen `paper_core` observations (all environments, dimensions and models pooled; 600 observations per intervention row) at the null-calibrated thresholds, and firing rate on the in-place near-null controls. Batch-level regimes use calibrated distributional sensors; item-aligned regimes use exact zero thresholds on item-wise deltas.

Intervention	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{Y_m}$	$\mathcal{I}_{XFY}$	$\mathcal{I}_{XF}$ item	$\mathcal{I}_{XFY}$ item
Feature sign flip 0.02	0.118	0.192	0.000	0.200	0.793	0.793
Feature sign flip 0.05	0.532	0.640	0.000	0.643	0.867	0.867
Feature sign flip 0.10	0.757	0.877	0.000	0.878	0.907	0.907
Mean shift 0.02	1.000	1.000	0.000	1.000	0.357	0.357
Mean shift 0.05	1.000	1.000	0.000	1.000	0.500	0.500
Mean shift 0.10	1.000	1.000	0.000	1.000	0.618	0.618
Scaling drift 0.02	1.000	1.000	0.000	1.000	0.537	0.537
Scaling drift 0.05	1.000	1.000	0.000	1.000	0.608	0.608
Scaling drift 0.10	1.000	1.000	0.000	1.000	0.668	0.668
Feature dropout 0.02	0.000	0.000	0.000	0.000	0.488	0.488
Feature dropout 0.05	0.000	0.022	0.000	0.022	0.705	0.705
Feature dropout 0.10	0.000	0.062	0.000	0.062	0.763	0.763
Prior-preserving label flip 0.02	0.000	0.000	0.000	0.000	0.000	1.000
Prior-preserving label flip 0.05	0.000	0.000	0.000	0.002	0.000	1.000
Prior-preserving label flip 0.10	0.000	0.000	0.000	0.033	0.000	1.000
Random label flip 0.02	0.000	0.000	0.000	0.000	0.000	1.000
Random label flip 0.05	0.000	0.000	0.000	0.003	0.000	1.000
Random label flip 0.10	0.000	0.000	0.000	0.033	0.000	1.000
Near-null Gaussian $\sigma = 0.001$	0.713	0.713	0.000	0.713	0.042	0.042
Near-null scaling $\alpha = 0.001$	0.879	0.882	0.000	0.882	0.099	0.099

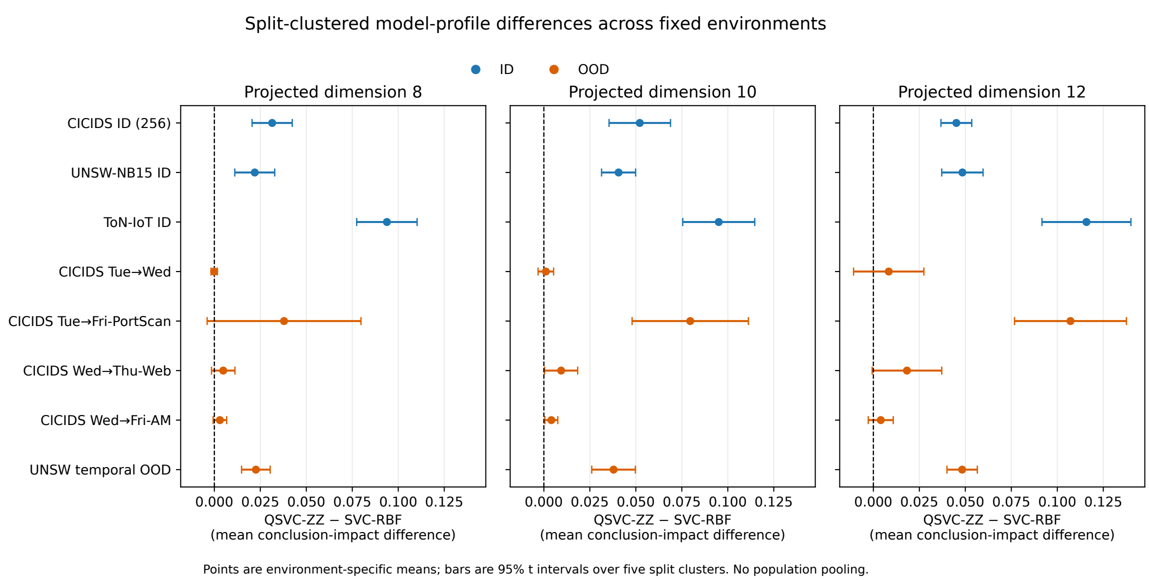

Fig. 1. Environment-specific ZZ-minus-SVC mean conclusion-impact differences and 95% t intervals over five split clusters. ID and OOD settings are fixed cases; no population pooling is performed.

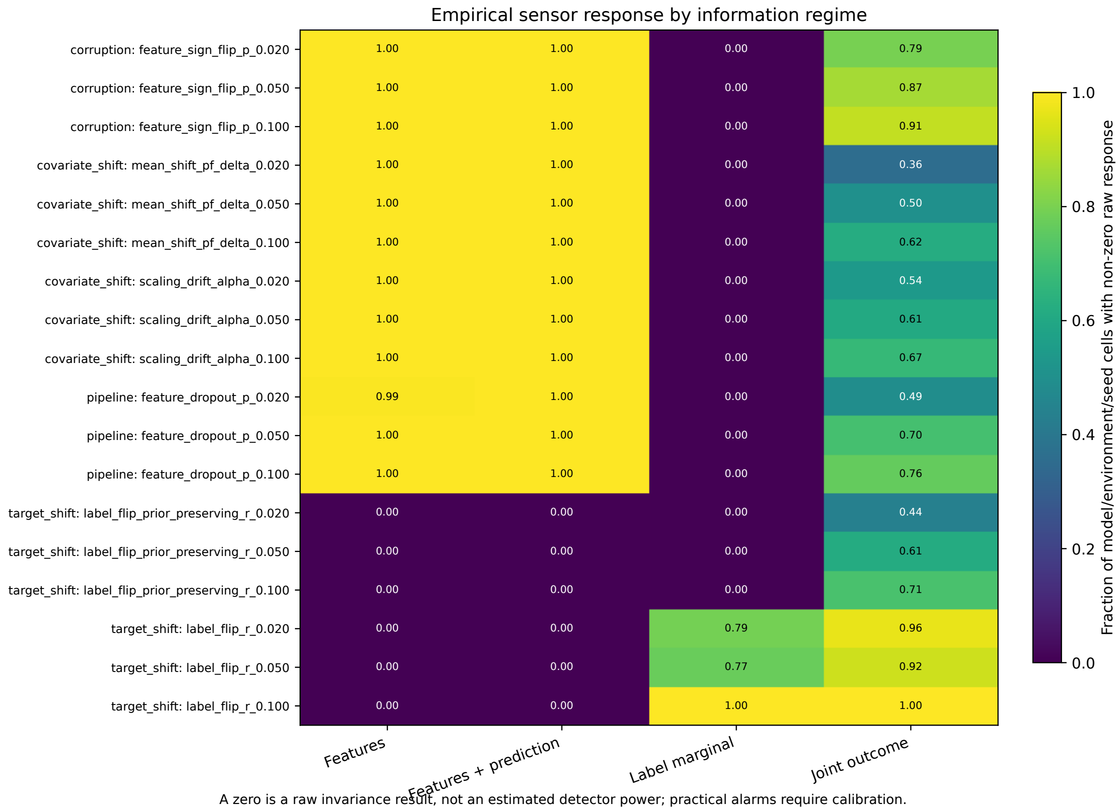

Fig. 2. Raw sensor response by information regime in the frozen expansion. Fractions are over model/environment/seed cells; zero denotes raw invariance, not estimated detector power.

TABLE 10
Gate N: label-path interventions in the frozen expansion; number of observations on which each regime fires after calibration (batch-level) or by exact item-aligned comparison.

Subset	n	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{Y_m}$	$\mathcal{I}_{XFY}$	item-aligned
All label interventions	3,600	0	0	0	43	3,600
Prior-preserving	1,800	0	0	0	21	1,800
Positive conclusion impact	2,184	0	0	0	40	2,184

TABLE 11
Cluster structure of the frozen evaluation batches (mean over the 30 cells of each environment, standardized projected features): fraction of near-duplicate rows, and the largest fraction of rows within a window of 0.01 or 0.001 standard deviations in any feature (max) or on average over features (mean).

Environment	Dup. rows	Max 0.01	Mean 0.01	Max 0.001
E1 CICIDS ID 256	0.074	0.825	0.562	0.648
E2 UNSW-NB15 ID	0.114	0.735	0.458	0.514
E3 ToN-IoT ID	0.257	0.870	0.466	0.571
E4 Tue→Wed	0.054	0.766	0.489	0.480
E5 Tue→Fri-PortScan	0.129	0.872	0.747	0.822
E6 Wed→Thu-Web	0.004	0.827	0.433	0.696
E7 Wed→Fri-AM	0.026	0.769	0.670	0.663
E8 UNSW temporal OOD	0.100	0.694	0.437	0.477

The authoritative counts of tests, manifests, outputs, files, pages and hashes for the released version are generated into `publication/RELEASE_STATUS.md` by `src/experiments/make_release_status.py`. The frozen environment uses Python 3.10.16, NumPy 2.2.6, pandas 2.3.3, scikit-learn 1.7.2, Qiskit 2.3.0 and Qiskit Machine Learning 0.9.0. Qiskit is pinned below 3 because migrating deprecated feature-map constructors would change circuit serialization and therefore requires a new evidence version. The artifact omits raw benchmark datasets; it documents their official sources, expected SHA-256 hashes, deterministic staging and a cross-platform hash verifier, and includes source code, locked dependencies, compact derived evidence, figures, figure-source tables, tests, manifests, preregistrations and expected outputs. Version 1.2.0 is archived at Zenodo under the concept DOI 10.5281/zenodo.22550852 with repository tag `paper15-q1-v1.2.0`; source code is licensed under Apache-2.0, derived evidence, figures and documentation under CC BY 4.0, and the manuscript files are author preprints excluded from both licenses.

14 Claims-Supported / Claims-Excluded Matrix

Supported by the artifact. Auditability is a property of (intervention class, view, trusted references); structural blind regions are monotone under refinement and closed exactly by added evidence that separates the class; feature/prediction sensors are exactly blind to label-only changes and label marginals to prior-preserving relabeling; every material label-path change alters the confusion matrix and is separable in the joint view, but batch-level auditors detect at most 1.6% of them whereas a trusted aggregate reference of the same batch detects all of them and an item-aligned reference additionally exposes the 808 aggregate-preserving relabelings; the blind

TABLE 12
Gates P and T: within-split ZZ-minus-SVC difference in suite-average balanced-accuracy conclusion impact, mean with 95% t interval over five split clusters and number of positive clusters. CICIDS ID reference is the re-executed P0 configuration; UNSW OOD reference is the frozen 1.0.0 environment E8.

Condition	$d = 8$	$d = 10$	$d = 12$
P0 reference (standard / minmax2π)	0.0245 [0.0069, 0.0422] (5/5)	0.0348 [0.0176, 0.0520] (5/5)	0.0407 [0.0305, 0.0509] (5/5)
PA standard / standard	0.0070 [0.0014, 0.0125] (5/5)	0.0130 [0.0041, 0.0219] (5/5)	0.0111 [0.0052, 0.0170] (5/5)
PB minmax2π / minmax2π	0.0238 [0.0054, 0.0423] (5/5)	0.0323 [0.0156, 0.0491] (5/5)	0.0396 [0.0261, 0.0531] (5/5)
CICIDS ID, both tuned (CV5)	0.0426 [0.0094, 0.0757] (5/5)	0.0713 [0.0494, 0.0932] (5/5)	0.0776 [0.0610, 0.0943] (5/5)
UNSW OOD, frozen defaults	0.0226 [0.0148, 0.0304] (5/5)	0.0379 [0.0260, 0.0498] (5/5)	0.0483 [0.0400, 0.0565] (5/5)
UNSW OOD, both tuned (CV5)	0.0229 [0.0103, 0.0356] (5/5)	0.0348 [0.0205, 0.0492] (5/5)	0.0459 [0.0371, 0.0546] (5/5)

TABLE 13
Gates P and T: clean balanced accuracy (SVC / ZZ, mean over five split clusters) per configuration and dimension, and the distribution of cross-validated regularisation constants over the 30 cells of each tuned environment (SVC also selects γ; see the artifact table).

Condition	$d = 8$	$d = 10$	$d = 12$
P0 reference (standard / minmax2π)	0.753 / 0.734	0.747 / 0.728	0.778 / 0.747
PA standard / standard	0.753 / 0.727	0.747 / 0.720	0.778 / 0.727
PB minmax2π / minmax2π	0.751 / 0.734	0.750 / 0.728	0.756 / 0.747
CICIDS ID, both tuned (CV5)	0.741 / 0.750	0.746 / 0.781	0.780 / 0.802
UNSW OOD, frozen defaults	0.748 / 0.760	0.754 / 0.766	0.755 / 0.769
UNSW OOD, both tuned (CV5)	0.739 / 0.759	0.723 / 0.757	0.736 / 0.770
Tuned environment	**QSVC C selections**		**SVC C selections**
CICIDS ID	$C = 0.1$: 3, $C = 1$: 4, $C = 10$: 6, $C = 100$: 17		$C = 0.1$: 1, $C = 1$: 6, $C = 10$: 15, $C = 100$: 8
UNSW OOD	$C = 1$: 17, $C = 10$: 13		$C = 0.1$: 1, $C = 1$: 2, $C = 10$: 15, $C = 100$: 12

TABLE 14
Gate F: decision-level false-alarm rate on the 12,000 disjoint clean evaluation draws (nominal $\alpha = 0.05$ per decision) of the union of per-sensor rules (1.1.0) and of the family-calibrated rule. Pooled rates are descriptive because draws within a cell overlap; the last column is the range of the eight per-environment cluster means (five split clusters each).

Regime	Rule	Fires	Pooled	Env. cluster means
$\mathcal{I}_X$	union	1,500	0.125	0.063–0.215
$\mathcal{I}_X$	family	733	0.061	0.025–0.133
$\mathcal{I}_{XF}$	union	2,438	0.203	0.135–0.280
$\mathcal{I}_{XF}$	family	871	0.073	0.045–0.125
$\mathcal{I}_{Y_m}$	union	570	0.048	0.025–0.070
$\mathcal{I}_{Y_m}$	family	570	0.048	0.025–0.070
$\mathcal{I}_{XFY}$	union	3,113	0.259	0.181–0.358
$\mathcal{I}_{XFY}$	family	942	0.079	0.046–0.146

regions persist under per-sensor and family-wise calibration; per-sensor calibration at $\alpha = 0.05$ yields decision false-alarm rates of 0.125–0.259 that family calibration reduces to 0.061–0.079 within the frozen design; batch-level regimes with feature evidence serve 4,322–4,494 of 7,008 materially changed results under calibrated policies, the label-marginal regime serves all of them, and the trusted item-aligned regime serves none; the secondary ZZ-minus-SVC profile is direction-stable but scaler- and headroom-dependent; complementary quantum-workflow sensors close different simulator-gate blind regions; and the local prototype executes policy-aware, fail-closed `allow/hold/block` composition.

Explicitly excluded. The evidence does not estimate attack prevalence or population-level deployment incidence, establish universal superiority or vulnerability of quantum kernels, causally isolate feature-map geometry from preprocessing and clean headroom, or establish that the exchangeability premise of the calibration holds in deployment. It provides no evidence about QPU security, calibrated device noise, scheduling, provider integrity or attestation, multi-tenancy, side channels, production readiness, non-repudiation, persistence, incident handling, recovery under non-stationarity, or operational Fleet Management.

TABLE 15

Gate F: per-environment false-alarm rate of the union rule (cluster mean) and of the family-calibrated rule (cluster mean with two-sided 95% t interval over the five split clusters, four degrees of freedom). n is the batch size; "disjoint" is the number of mutually disjoint batches that one pool half can supply, an upper bound on the effective number of independent draws among the 20 per run (E1: one).

Environment	n	disj.	$\mathcal{I}_X$: union, family [95% CI]	$\mathcal{I}_{XF}$: union, family [95% CI]	$\mathcal{I}_{Y_m}$: union, family [95% CI]	$\mathcal{I}_{XFY}$: union, family [95% CI]
E1 CICIDS ID 256	256	1	0.203 0.133 [-0.05, 0.31]	0.245 0.125 [-0.02, 0.27]	0.045 0.045 [-0.05, 0.14]	0.331 0.146 [0.00, 0.29]
E2 UNSW-NB15 ID	128	3	0.063 0.025 [-0.02, 0.07]	0.135 0.051 [0.01, 0.09]	0.070 0.070 [0.01, 0.13]	0.210 0.054 [0.01, 0.10]
E3 ToN-IoT ID	128	3	0.215 0.071 [-0.03, 0.17]	0.280 0.088 [0.01, 0.17]	0.070 0.070 [0.01, 0.13]	0.358 0.102 [-0.00, 0.20]
E4 Tue→Wed	128	11	0.150 0.098 [-0.01, 0.20]	0.230 0.095 [0.02, 0.17]	0.025 0.025 [-0.01, 0.06]	0.254 0.102 [0.02, 0.18]
E5 Tue→Fri-PortScan	128	11	0.066 0.039 [0.00, 0.08]	0.160 0.045 [-0.01, 0.10]	0.025 0.025 [-0.01, 0.06]	0.197 0.046 [-0.00, 0.09]
E6 Wed→Thu-Web	128	11	0.070 0.044 [0.00, 0.09]	0.163 0.054 [-0.00, 0.11]	0.025 0.025 [-0.01, 0.06]	0.181 0.054 [0.01, 0.10]
E7 Wed→Fri-AM	128	11	0.133 0.060 [0.02, 0.10]	0.212 0.063 [0.04, 0.09]	0.025 0.025 [-0.01, 0.06]	0.257 0.061 [0.04, 0.09]
E8 UNSW temporal OOD	128	11	0.072 0.043 [-0.03, 0.12]	0.192 0.065 [0.02, 0.11]	0.050 0.050 [-0.01, 0.11]	0.237 0.065 [0.03, 0.10]

TABLE 16

Gate F: detection rate on the frozen `paper_core` observations (600 per intervention row, eight environments pooled) and firing rate on the near-null in-place controls, as union rule / family-calibrated rule. The family rule trades a small loss of detection for the decision-level false-alarm budget of Table 14.

Intervention	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{Y_m}$	$\mathcal{I}_{XFY}$
Feature sign flip 0.02	0.12 / 0.09	0.19 / 0.15	0.00 / 0.00	0.20 / 0.15
Feature sign flip 0.05	0.53 / 0.47	0.64 / 0.56	0.00 / 0.00	0.64 / 0.53
Feature sign flip 0.10	0.76 / 0.72	0.88 / 0.85	0.00 / 0.00	0.88 / 0.84
Mean shift 0.02	1.00 / 1.00	1.00 / 1.00	0.00 / 0.00	1.00 / 0.99
Mean shift 0.05	1.00 / 1.00	1.00 / 1.00	0.00 / 0.00	1.00 / 1.00
Mean shift 0.10	1.00 / 1.00	1.00 / 1.00	0.00 / 0.00	1.00 / 1.00
Scaling drift 0.02	1.00 / 0.99	1.00 / 0.99	0.00 / 0.00	1.00 / 0.98
Scaling drift 0.05	1.00 / 1.00	1.00 / 1.00	0.00 / 0.00	1.00 / 1.00
Scaling drift 0.10	1.00 / 1.00	1.00 / 1.00	0.00 / 0.00	1.00 / 1.00
Feature dropout 0.02	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
Feature dropout 0.05	0.00 / 0.00	0.02 / 0.01	0.00 / 0.00	0.02 / 0.01
Feature dropout 0.10	0.00 / 0.00	0.06 / 0.04	0.00 / 0.00	0.06 / 0.03
Prior-preserving label flip 0.02	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
Prior-preserving label flip 0.05	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
Prior-preserving label flip 0.10	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00	0.03 / 0.01
Random label flip 0.02	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
Random label flip 0.05	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
Random label flip 0.10	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00	0.03 / 0.01
Near-null Gaussian $\sigma = 0.001$	0.65 / 0.49	0.65 / 0.44	0.00 / 0.00	0.65 / 0.43
Near-null scaling $\alpha = 0.001$	0.86 / 0.68	0.86 / 0.64	0.00 / 0.00	0.86 / 0.62

TABLE 17

Gate D: complete decision counts at the primary materiality threshold ($|\Delta_R| > 0$; 7,008 material and 3,792 non-material intervened observations; 1,200 near-null benign shams). P0 serve-always, P1 union, P2 family-calibrated, P3 strict. Clean draws: 12,000 for batch-level regimes; the 1,200 exact-zero rows for $\mathcal{I}^{\star}_{XFY}$. Ben. h/b: fraction of benign shams held or blocked. Integ.: non-material intervened observations held or blocked. Safe allow: clean, benign and non-material observations allowed.

Regime	Pol.	Clean	F. hold	F. block	Ben. h/b	Unsafe	T. hold	T. block	Integ.	Imm. allow	Safe allow
$\mathcal{I}_X$	P0	12,000	0	0	0.00	7,008	0	0	0	3,792	16,992
$\mathcal{I}_X$	P1	12,000	1,500	0	0.76	4,429	2,579	0	1,865	1,927	12,720
$\mathcal{I}_X$	P2	12,000	733	0	0.58	4,494	2,514	0	1,851	1,941	13,706
$\mathcal{I}_X$	P3	12,000	12,000	0	1.00	0	7,008	0	3,792	0	0
$\mathcal{I}_{XF}$	P0	12,000	0	0	0.00	7,008	0	0	0	3,792	16,992
$\mathcal{I}_{XF}$	P1	12,000	2,438	0	0.76	4,231	2,777	0	1,898	1,894	11,746
$\mathcal{I}_{XF}$	P2	12,000	871	0	0.54	4,327	2,681	0	1,873	1,919	13,600
$\mathcal{I}_{XF}$	P3	12,000	12,000	0	1.00	0	7,008	0	3,792	0	0
$\mathcal{I}_{Y_m}$	P0	12,000	0	0	0.00	7,008	0	0	0	3,792	16,992
$\mathcal{I}_{Y_m}$	P1	12,000	570	0	0.00	7,008	0	0	0	3,792	16,422
$\mathcal{I}_{Y_m}$	P2	12,000	570	0	0.00	7,008	0	0	0	3,792	16,422
$\mathcal{I}_{Y_m}$	P3	12,000	12,000	0	1.00	0	7,008	0	3,792	0	0
$\mathcal{I}_{XFY}$	P0	12,000	0	0	0.00	7,008	0	0	0	3,792	16,992
$\mathcal{I}_{XFY}$	P1	12,000	3,113	0	0.76	4,180	2,828	0	1,898	1,894	11,071
$\mathcal{I}_{XFY}$	P2	12,000	942	0	0.53	4,322	2,686	0	1,859	1,933	13,561
$\mathcal{I}_{XFY}$	P3	12,000	942	0	0.53	4,322	2,686	0	1,859	1,933	13,561
$\mathcal{I}^{\star}_{XFY}$	P0	1,200	0	0	0.00	7,008	0	0	0	3,792	6,192
$\mathcal{I}^{\star}_{XFY}$	P1	1,200	0	0	0.77	0	0	7,008	3,016	776	2,250
$\mathcal{I}^{\star}_{XFY}$	P2	1,200	0	0	0.55	0	0	7,008	2,981	811	2,546
$\mathcal{I}^{\star}_{XFY}$	P3	1,200	0	0	0.55	0	0	7,008	2,981	811	2,546

TABLE 18
Gate D: unsafe allows (containment in parentheses) under three materiality thresholds on $|\Delta_R|$; material observations 7,008, 3,802 and 1,870. P1 union, P2 family-calibrated.

Regime	Pol.	$\tau = 0^+$	$\tau = 0.02$	$\tau = 0.05$
$\mathcal{I}_X$	P1	4,429 (0.37)	2,128 (0.44)	652 (0.65)
$\mathcal{I}_X$	P2	4,494 (0.36)	2,174 (0.43)	680 (0.64)
$\mathcal{I}_{XF}$	P1	4,231 (0.40)	1,977 (0.48)	566 (0.70)
$\mathcal{I}_{XF}$	P2	4,327 (0.38)	2,046 (0.46)	609 (0.67)
$\mathcal{I}_{Y_m}$	P1	7,008 (0.00)	3,802 (0.00)	1,870 (0.00)
$\mathcal{I}_{Y_m}$	P2	7,008 (0.00)	3,802 (0.00)	1,870 (0.00)
$\mathcal{I}_{XFY}$	P1	4,180 (0.40)	1,926 (0.49)	521 (0.72)
$\mathcal{I}_{XFY}$	P2	4,322 (0.38)	2,036 (0.46)	597 (0.68)
$\mathcal{I}^\star_{XFY}$	P1	0 (1.00)	0 (1.00)	0 (1.00)
$\mathcal{I}^\star_{XFY}$	P2	0 (1.00)	0 (1.00)	0 (1.00)

TABLE 19
Gate D: unsafe allows / material observations by intervention mechanism under the family-calibrated policy P2 ($|\Delta_R| > 0$).

Regime	Evaluation-label	Sign flip	Mean shift / scaling	Feature dropout
$\mathcal{I}_X$	2,617/2,617	798/1,464	0/1,848	1,079/1,079
$\mathcal{I}_{XF}$	2,617/2,617	655/1,464	2/1,848	1,053/1,079
$\mathcal{I}_{Y_m}$	2,617/2,617	1,464/1,464	1,848/1,848	1,079/1,079
$\mathcal{I}_{XFY}$	2,601/2,617	663/1,464	3/1,848	1,055/1,079
$\mathcal{I}^\star_{XFY}$	0/2,617	0/1,464	0/1,848	0/1,079